

Ejercicio 1.

Multiplicar los siguientes números y utilizar las técnicas de reducción de árboles de suma Carry Save Adder, Dual Carry Save Adder, Wallace Tree y Dadda Tree para optimizar el número de sumadores:

$$\begin{array}{l} 10_{-}1101 \rightarrow (45) \\ 11_{-}1011 \rightarrow (59) \end{array} \Rightarrow 45 \times 59 = 2655$$

Para todos los casos

1. Mostrar el paso a paso de las operaciones de suma en el multiplicador.
2. Dibujar los diagramas en bloques de la arquitectura obtenida al aplicar cada método de reducción de complejidad.
3. Calcular el número de FA y HA utilizados.

Carry Save Reduction

$$\begin{array}{r} 101101 \\ 111011 \\ \hline 101101 \\ 101101 \\ 000000 \\ 101101 \\ 101101 \\ 101101 \\ \hline \end{array} \quad \begin{array}{l} PP_0 \\ PP_1 \\ PP_2 \\ PP_3 \\ PP_4 \\ PP_5 \end{array} \quad \left. \vphantom{\begin{array}{l} PP_0 \\ PP_1 \\ PP_2 \\ PP_3 \\ PP_4 \\ PP_5 \end{array}} \right\} CSR$$

Estep 1 (PP₀, PP₁, PP₂)

$$\begin{array}{r} + \quad 00000101101 \\ + \quad 00000101101 \\ + \quad 00000000000 \\ \hline 000001110111 \rightarrow S_1 \\ 000000010000 \rightarrow C_1 \end{array}$$

Etape 2 (S_1, L_1, PP_3)

$$\begin{array}{r}
 + \begin{array}{cccccccccccc} 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{array} \\
 + \begin{array}{cccccccccccc} 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{array} \\
 \hline
 \begin{array}{cccccccccccc} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \end{array} \begin{array}{l} \longrightarrow S_2 \\ \longrightarrow L_2 \end{array}
 \end{array}$$

Etape 3 (S_2, L_2, PP_4)

$$\begin{array}{r}
 + \begin{array}{cccccccccccc} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \end{array} \\
 + \begin{array}{cccccccccccc} 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \end{array} \\
 \hline
 \begin{array}{cccccccccccc} 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \begin{array}{l} \longrightarrow S_3 \\ \longrightarrow L_3 \end{array}
 \end{array}$$

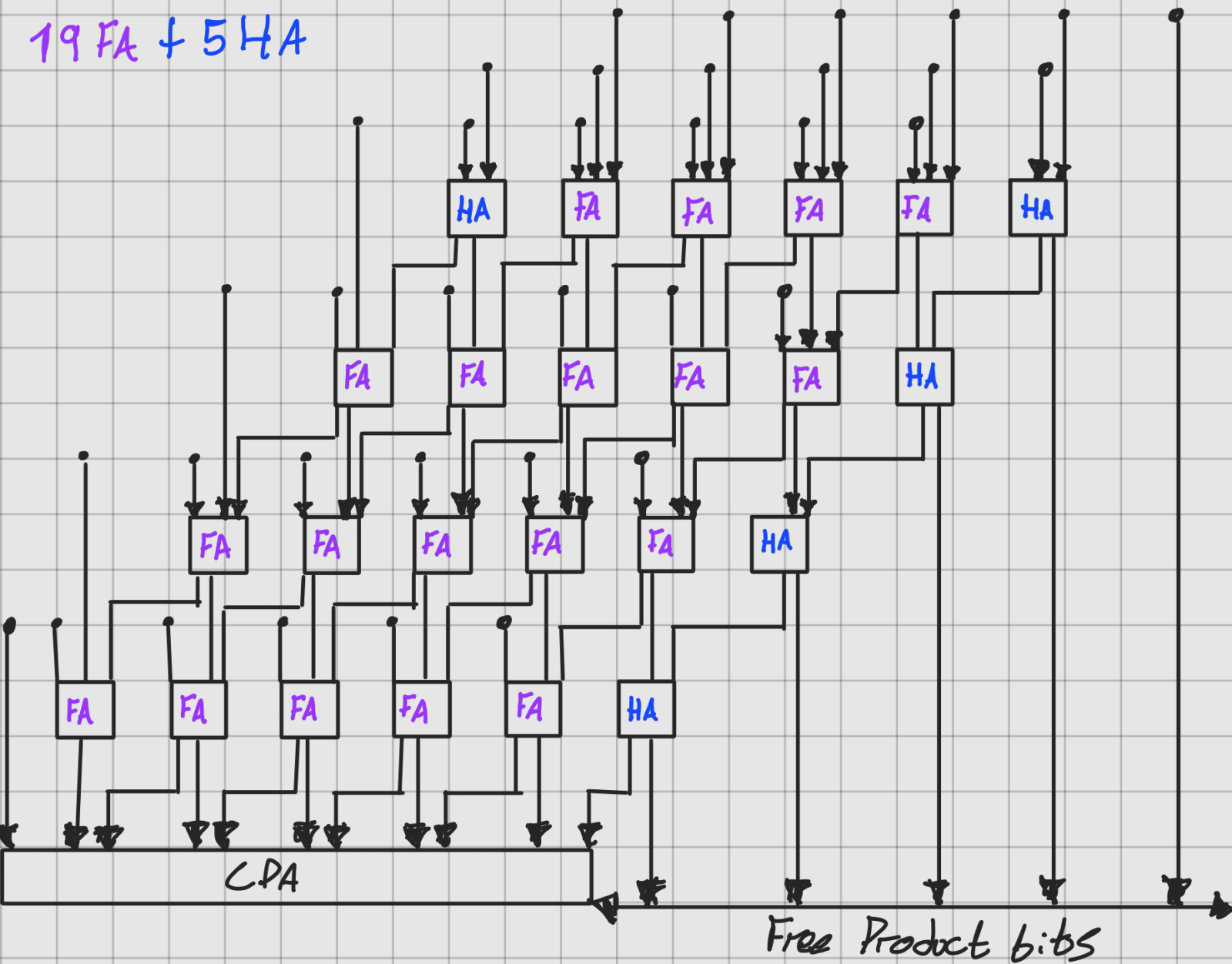
Etape 4 (S_3, L_3, PP_5)

$$\begin{array}{r}
 + \begin{array}{cccccccccccc} 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \\
 + \begin{array}{cccccccccccc} 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{array} \\
 \hline
 \begin{array}{cccccccccccc} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \begin{array}{l} \longrightarrow S_4 \\ \longrightarrow L_4 \end{array}
 \end{array}$$

Sum Final (CPA)

$$\begin{array}{r}
 + \begin{array}{cccccccccccc} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \\
 \hline
 \begin{array}{cccccccccccc} 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \end{array} = 2655_{10}
 \end{array}$$

19 FA + 5 HA



Dual Carry Save Reduction

$$\begin{array}{r}
 101101 \\
 111011 \\
 \hline
 101101 \\
 101101 \\
 000000 \\
 101101 \\
 101101 \\
 101101 \\
 \hline
 \end{array}
 \quad
 \begin{array}{l}
 PP_0 \\
 PP_1 \\
 PP_2 \\
 PP_3 \\
 PP_4 \\
 PP_5
 \end{array}
 \left. \vphantom{\begin{array}{l} PP_0 \\ PP_1 \\ PP_2 \\ PP_3 \\ PP_4 \\ PP_5 \end{array}} \right\} CSR1
 \quad
 \left. \vphantom{\begin{array}{l} PP_3 \\ PP_4 \\ PP_5 \end{array}} \right\} CSR2$$

Steps 1 (PP_0, PP_1, PP_2)

$$\begin{array}{r}
 + \quad 00000101101 \\
 + \quad 000001011010 \\
 + \quad 000000000000 \\
 \hline
 000001110111 \longrightarrow S_{13} \\
 000000010000 \longrightarrow C_{13}
 \end{array}$$

Steps 2 (PP_3, PP_4, PP_5)

$$\begin{array}{r}
 + \quad 000101101000 \\
 + \quad 001011010000 \\
 + \quad 010110100000 \\
 \hline
 011000011000 \longrightarrow S_{16} \\
 001111000000 \longrightarrow C_{16}
 \end{array}$$

Etapa 3 (S_{12}, C_{12}, S_{16})

$$\begin{array}{r}
 + \begin{array}{cccccccccccc} 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \end{array} \\
 + \begin{array}{cccccccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{array} \\
 + \begin{array}{cccccccccccc} 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \end{array} \\
 \hline
 \begin{array}{cccccccccccc} 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{array} \longrightarrow S_2 \\
 \begin{array}{cccccccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{array} \longrightarrow C_2
 \end{array}$$

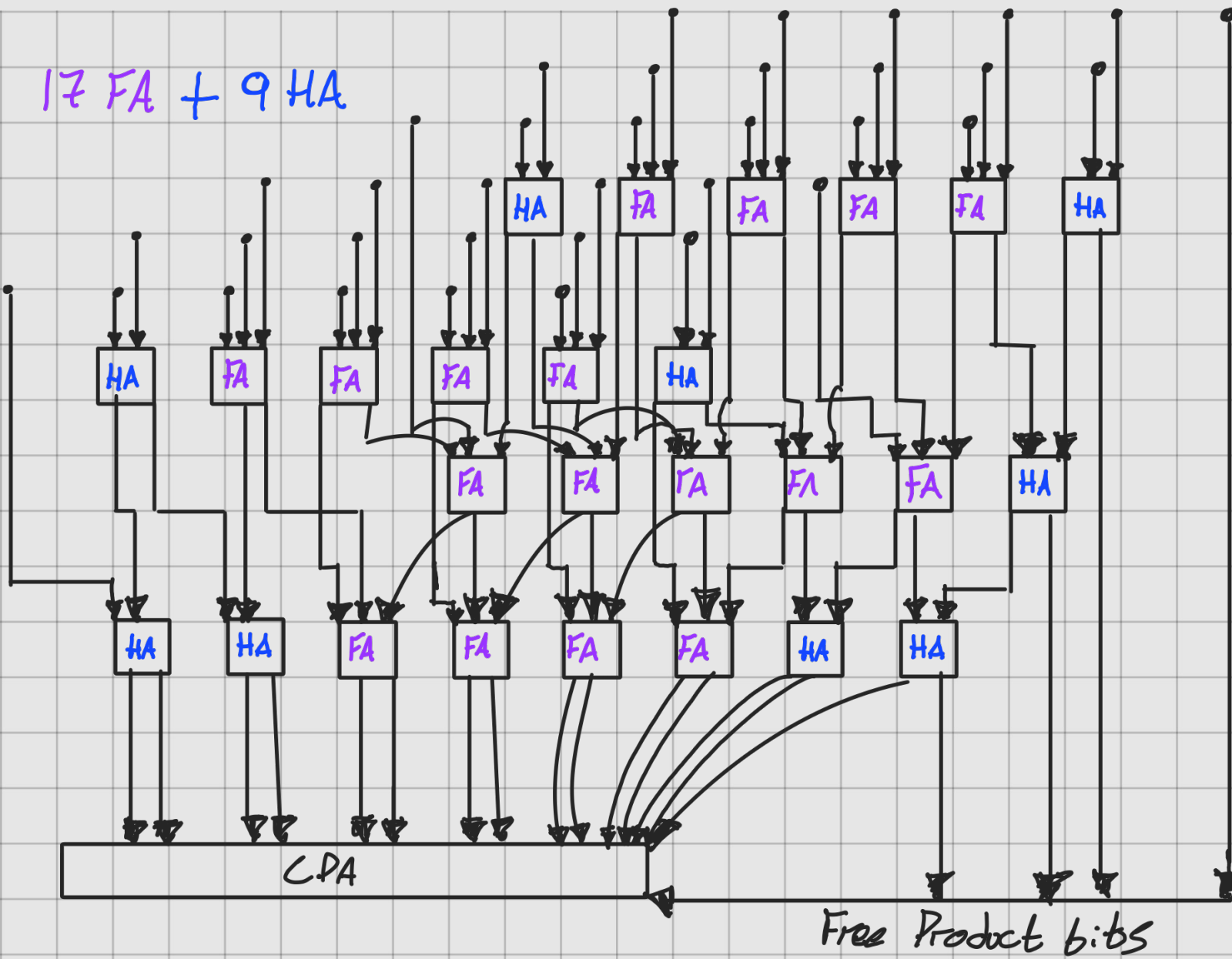
Etapa 4 (S_2, C_2, C_{16})

$$\begin{array}{r}
 + \begin{array}{cccccccccccc} 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{array} \\
 + \begin{array}{cccccccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{array} \\
 + \begin{array}{cccccccccccc} 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \\
 \hline
 \begin{array}{cccccccccccc} 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{array} \longrightarrow S_3 \\
 \begin{array}{cccccccccccc} 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \longrightarrow C_3
 \end{array}$$

Soma Final (CPA)

$$\begin{array}{cccccccccccc}
 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\
 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 \hline
 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 1
 \end{array}
 = 2655_{10}$$

17 FA + 9 HA



Wallace Tree

Level 1

1	0	1	1	0	1	
1	1	1	0	1	1	
	1	0	1	1	0	1
1	0	1	1	0	1	

PP_0
 PP_1
 PP_2
 PP_3
 PP_4
 PP_5

	0	0	0	0	0	0
1	0	1	1	0	1	

1	0	1	1	0	1
---	---	---	---	---	---

1 0 1 1 0 1

Level 2

$$\left. \begin{array}{l} S_{12} \\ C_{12} \\ S_{16} \\ C_{16} \end{array} \right\} CSRZ$$

Level 3

$$\left. \begin{array}{l} 52 \\ C2 \\ C96 \end{array} \right\} CSR3$$

S_3 $\begin{Bmatrix} 7 \\ 3 \end{Bmatrix}$ CPA

$\Rightarrow$ El procedimiento es el mismo que para CSA reduction para mult. 6×6

Dabo Tree

A_5	A_4	A_3	A_2	A_1	A_0
B_5	B_4	B_3	B_2	B_1	B_0

$$A_5B_0, A_4B_1, A_3B_2, A_2B_3, A_1B_4, A_0B_5$$
$$A_5B_1, A_4B_1, A_3B_1, A_2B_1, A_1B_1, A_0B_1$$
$$A_5B_2 \quad A_4B_2 \quad A_3B_2 \quad A_2B_2 \quad A_1B_2 \quad A_0B_2$$
$$A_5B_3, A_4B_3, A_3B_3, A_2B_3, A_1B_3, A_0B_3$$
$$A_5B_4, A_4B_4, A_3B_4, A_2B_4, A_1B_4, A_0B_4$$
$$A_5B_5, A_4B_5, A_3B_5, A_2B_5, A_1B_5, A_0B_5$$
$$A_5B_5, A_5B_4, A_5B_3, A_5B_2, A_5B_1, A_5B_0, A_4B_6, A_4B_5, A_4B_4, A_4B_3, A_4B_2, A_4B_1, A_4B_0, A_3B_7, A_3B_6, A_3B_5, A_3B_4, A_3B_3, A_3B_2, A_3B_1, A_3B_0, A_2B_8, A_2B_7, A_2B_6, A_2B_5, A_2B_4, A_2B_3, A_2B_2, A_2B_1, A_2B_0, A_1B_9, A_1B_8, A_1B_7, A_1B_6, A_1B_5, A_1B_4, A_1B_3, A_1B_2, A_1B_1, A_1B_0, A_0B_{10}, A_0B_9, A_0B_8, A_0B_7, A_0B_6, A_0B_5, A_0B_4, A_0B_3, A_0B_2, A_0B_1, A_0B_0$$
$$A_1B_5, A_1B_4, A_1B_3, A_1B_2, A_1B_1, A_2B_5, A_2B_4, A_2B_3, A_2B_2, A_2B_1, A_3B_5, A_3B_4, A_3B_3, A_3B_2, A_3B_1, A_4B_5, A_4B_4, A_4B_3, A_4B_2, A_4B_1, A_5B_5, A_5B_4, A_5B_3, A_5B_2, A_5B_1$$
$$A_3B_3, A_3B_4, A_3B_5, A_3B_6, A_2B_7, A_1B_8, A_0B_9$$
$$A_2B_3, A_2B_2, A_2B, AB_3, AB_2$$

A_1B_1 A_1B_2 A_2B_1 A_2B_2

AB

$$A_5 B_5 \quad A_5 B_4 \quad [A_5 B_3 \quad A_5 B_2] \quad [S_6 A \quad S_5 A \quad S_4 A] \quad [A_3 B_0 \quad A_2 B_0 \quad A_1 B_0 \quad A_0 B_0]$$
$$A_4 B_5, A_4 B_6, C_6 A, C_5 A, C_4 A, A_2 B_2, A_2 B_3, A_1 B_1, A_0 B_1$$
$$A_3B_8 \quad S_7A \quad S_6B \quad S_{5B} \quad A_1B_3 \quad A_1B_2 \quad A_0B_2$$

C7A C6B C5B A₀B₃ A₁B₄ A₂B₅

$A_5B_5, A_5B_4, S_6A, S_7B, S_6C, S_5C, S_4B, S_3A, A_7B_0, A_6B_0, A_0B_0$

$A_4 B_5, C_7 B, C_6 C, C_5 C, C_4 B, C_3 A, A_1 B_2, A_1 B_1, A_0 B_1$

$C_{6A} \quad C_{7A} \quad C_{6B} \quad C_{5B} \quad A_B \quad A_B \quad A_B \quad A_B$

$A_5 B_5$ S_{50} S_{51} S_{52} S_{53} S_{54} S_{55} S_{56} S_{57} S_{58} S_{59} S_{60} S_{61} S_{62} S_{63} S_{64} S_{65} S_{66} S_{67} S_{68} S_{69} S_{70} S_{71} S_{72} S_{73} S_{74} S_{75} S_{76} S_{77} S_{78} S_{79} S_{80} S_{81} S_{82} S_{83} S_{84} S_{85} S_{86} S_{87} S_{88} S_{89} S_{90} S_{91} S_{92} S_{93} S_{94} S_{95} S_{96} S_{97} S_{98} S_{99} S_{100}

C9A C8B C7C C6D C5B C4C C3B C2A AB AB

F:ndl $\angle PA = 70^\circ$ B:ls

PPD

PP₁

$$P_2$$
 PP_3

PP4

 PP_5

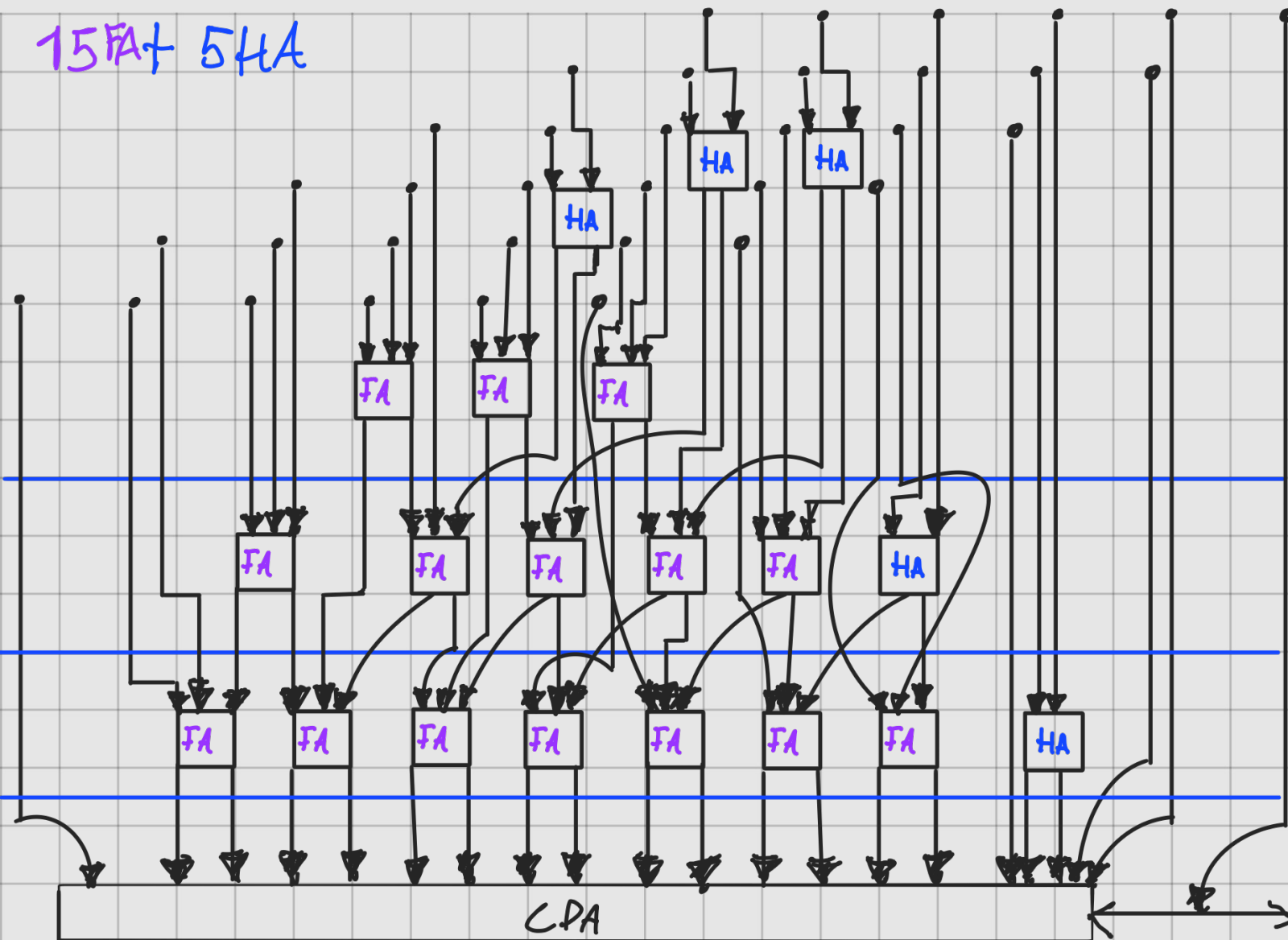
1st etape
(3FA + 3HA)

Zdaj etp
(5 FA + 7 HA)

3rd steps
(7FA + 1HA)

$$\text{Total} = 15FA + 5HA$$

15 FA + 5 HA



$$\begin{array}{r}
 101101 \\
 + 111011 \\
 \hline
 111011101 \\
 00001101 \\
 1110000 \\
 11101 \\
 001 \\
 1 \\
 \hline
 11101111101 \\
 00000101 \\
 1001000 \\
 110111 \\
 \hline
 11001010101 \\
 000101001 \\
 11101110 \\
 \hline
 1011011101 \\
 100101001 \\
 \hline
 10100101111 = 2655_{10}
 \end{array}$$